

# Representative Structural Calculation Package

Open parking frame

| Item            | Value                                                        |
|-----------------|--------------------------------------------------------------|
| Project         | Open parking frame                                           |
| Purpose         | Representative elastic analysis and preliminary design trace |
| Generated       | 2026-06-27T00:00:00.000Z                                     |
| Package version | m42-calculation-package                                      |
| Source model    | 06-open-parking-frame                                        |

|       |         |       |        |           |           |
|-------|---------|-------|--------|-----------|-----------|
| Nodes | Members | Loads | Combos | Max disp. | Max util. |
| 72    | 145     | 410   | 28     | 13.55 mm  | 1.045     |

## Model And Design Basis

| Item            | Value                                                        |
|-----------------|--------------------------------------------------------------|
| Building type   | parking                                                      |
| Description     | Five-story open parking structure with repetitive long bays. |
| Occupancy       | Parking                                                      |
| Bounds X/Y/Z    | 22.50 / 15.00 / 16.00                                        |
| Analysis status | OK                                                           |
| Design status   | OK                                                           |
| Package audit   | OK                                                           |

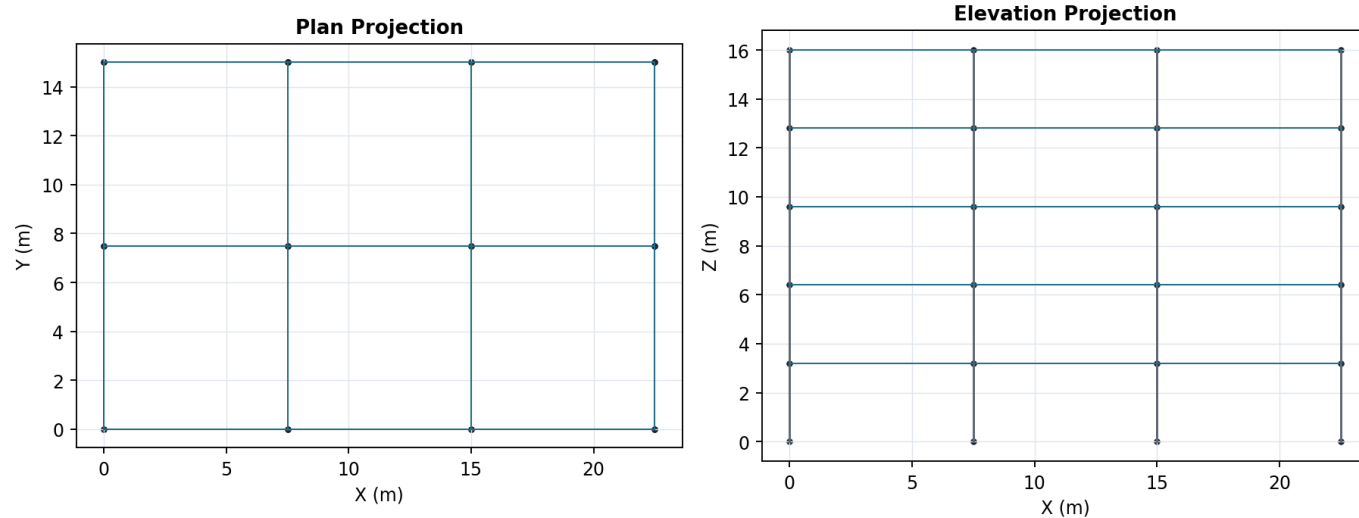
## Load Case Trace

| Case | Type    | Loads | Fx        | Fy        | Fz       |
|------|---------|-------|-----------|-----------|----------|
| D    | dead    | 85    | 0.000 kN  | 0.000 kN  | -5400 kN |
| L    | live    | 85    | 0.000 kN  | 0.000 kN  | -2734 kN |
| WX   | wind    | 60    | 108.00 kN | 0.000 kN  | 0.000 kN |
| WY   | wind    | 60    | 0.000 kN  | 162.00 kN | 0.000 kN |
| EX   | seismic | 60    | 517.09 kN | 0.000 kN  | 0.000 kN |
| EY   | seismic | 60    | 0.000 kN  | 517.09 kN | 0.000 kN |

## Load Derivation

| Story | Area   | D total | L total   | Wind X   | Wind Y   |
|-------|--------|---------|-----------|----------|----------|
| 1     | 337.50 | 1080 kN | 607.50 kN | 21.60 kN | 32.40 kN |
| 2     | 337.50 | 1080 kN | 607.50 kN | 21.60 kN | 32.40 kN |
| 3     | 337.50 | 1080 kN | 607.50 kN | 21.60 kN | 32.40 kN |
| 4     | 337.50 | 1080 kN | 607.50 kN | 21.60 kN | 32.40 kN |
| 5     | 337.50 | 1080 kN | 303.75 kN | 21.60 kN | 32.40 kN |

### Open parking frame



## Combination Analysis Results

| Combo          | Status | Max disp. | Max util. | Residual |
|----------------|--------|-----------|-----------|----------|
| KDS-ST-01      | OK     | 2.540 mm  | 0.235     | 0.000    |
| KDS-ST-02      | OK     | 3.445 mm  | 0.337     | 0.000    |
| KDS-ST-04-WX-P | OK     | 3.545 mm  | 0.330     | 0.000    |
| KDS-ST-04-WX-N | OK     | 3.545 mm  | 0.330     | 0.000    |
| KDS-ST-04-WY-P | OK     | 4.395 mm  | 0.356     | 0.000    |
| KDS-ST-04-WY-N | OK     | 4.395 mm  | 0.356     | 0.000    |
| KDS-ST-05-EX-P | OK     | 12.53 mm  | 0.505     | 0.000    |
| KDS-ST-05-EX-N | OK     | 12.53 mm  | 0.505     | 0.000    |
| KDS-ST-05-EY-P | OK     | 13.55 mm  | 0.518     | 0.000    |
| KDS-ST-05-EY-N | OK     | 13.55 mm  | 0.518     | 0.000    |
| KDS-ST-06-WX-P | OK     | 2.627 mm  | 0.195     | 0.000    |
| KDS-ST-06-WX-N | OK     | 2.627 mm  | 0.195     | 0.000    |
| KDS-ST-06-WY-P | OK     | 3.703 mm  | 0.221     | 0.000    |
| KDS-ST-06-WY-N | OK     | 3.703 mm  | 0.221     | 0.000    |
| KDS-ST-07-EX-P | OK     | 12.29 mm  | 0.371     | 0.000    |
| KDS-ST-07-EX-N | OK     | 12.29 mm  | 0.371     | 0.000    |
| KDS-ST-07-EY-P | OK     | 13.34 mm  | 0.383     | 0.000    |

|                |    |          |       |       |
|----------------|----|----------|-------|-------|
| KDS-ST-07-EY-N | OK | 13.34 mm | 0.383 | 0.000 |
|----------------|----|----------|-------|-------|

## Governing Members

| Rank | Member | Status | Util. | Check            | Combo          |
|------|--------|--------|-------|------------------|----------------|
| 1    | M55    | NG     | 1.045 | steel-deflection | KDS-ST-05-EY-N |
| 2    | M54    | NG     | 1.045 | steel-deflection | KDS-ST-05-EY-P |
| 3    | M59    | NG     | 1.042 | steel-deflection | KDS-ST-05-EX-N |
| 4    | M51    | NG     | 1.042 | steel-deflection | KDS-ST-05-EX-N |
| 5    | M58    | NG     | 1.042 | steel-deflection | KDS-ST-05-EX-P |
| 6    | M50    | NG     | 1.042 | steel-deflection | KDS-ST-05-EX-P |
| 7    | M56    | NG     | 1.040 | steel-deflection | KDS-ST-05-EY-N |
| 8    | M53    | NG     | 1.040 | steel-deflection | KDS-ST-05-EY-N |
| 9    | M60    | NG     | 1.038 | steel-deflection | KDS-ST-05-EY-P |
| 10   | M52    | NG     | 1.038 | steel-deflection | KDS-ST-05-EY-N |
| 11   | M49    | NG     | 1.038 | steel-deflection | KDS-ST-05-EY-N |
| 12   | M57    | NG     | 1.038 | steel-deflection | KDS-ST-05-EY-P |
| 13   | M43    | WARN   | 0.926 | steel-deflection | KDS-ST-05-EY-N |
| 14   | M42    | WARN   | 0.926 | steel-deflection | KDS-ST-05-EY-N |
| 15   | M39    | WARN   | 0.922 | steel-deflection | KDS-ST-05-EX-P |
| 16   | M47    | WARN   | 0.922 | steel-deflection | KDS-ST-05-EX-P |
| 17   | M38    | WARN   | 0.922 | steel-deflection | KDS-ST-05-EX-N |
| 18   | M46    | WARN   | 0.922 | steel-deflection | KDS-ST-05-EX-N |
| 19   | M44    | WARN   | 0.920 | steel-deflection | KDS-ST-05-EY-P |
| 20   | M41    | WARN   | 0.920 | steel-deflection | KDS-ST-05-EY-P |

## Steel, Connection, Foundation Summary

| Scope        | Rows | Warnings/NG | Max ratio |
|--------------|------|-------------|-----------|
| Steel review | 145  | 40          | 1.045     |
| Connections  | 145  | 50          | 5.206     |
| Foundations  | 12   | 50          | 1.000     |

## Quality Audit

| Audit item                    | Status         |
|-------------------------------|----------------|
| Load derivation attached      | OK             |
| Combination results available | OK             |
| Member checks available       | OK             |
| RC schedule available         | Not applicable |
| Steel schedule available      | OK             |
| Foundation review available   | OK             |

## Remaining Scope

- Project design basis: occupancy, importance factor, site class, exposure, wind/seismic procedure, and serviceability criteria.
- Load derivation: dead, live, wind, seismic, snow, rain, soil, temperature, and construction load calculation sheets.
- Full KDS equation trace for every generated load combination and project-specific exception.

- RC member detailing: longitudinal bars, stirrups, development length, lap splice, anchorage, and constructability checks.
- Steel member detailing: compactness, lateral torsional buckling, connection forces, base plates, and weld/bolt checks.
- Foundation design: soil bearing, pile/footing design, settlement, uplift, sliding, overturning, and reinforcement.
- Drawing import and vision-assisted modeling audit trail for future CAD/image/MGT workflows.
- Drawing/image/MGT import audit trail is not attached to this generated test set.